

PG Department of Electronics
N S S College, Rajakumari

PROJECT PROPOSAL



February, 2019

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1. Summary

Department of Electronics came into existence at N S S College, Rajakumari since the inception of the college on 11th of September, 1995. During the last 22 years, department has moulded many technocrats, academicians, researchers, professionals with expertise in software and hardware arena. During the year 2014-15 department is elevated to Post Graduate Department with the commencement of MSc Electronics Post Graduate Programme.

The laboratory sessions in the curriculum aims to give basic training on theories in the textbook and lacks professional training in design, development and implementation of a project. To attain additional skill to meet the requirement of the industry, sessions have to be conducted to train the students in the latest and current trends in the electronics and allied area.

As per the new power requirement, our nation has the plans to build renewable energy capacity of 175 GW by 2022. India's per capita electricity consumption, though increasing, is lowest among the BRICS nations and about one-third the world's average. The energy deficiency is a low 2.1%, but experts feel latent demand from remote areas is not being accounted for. Demand for electricity is likely to pick up after 2019 as the scheme to revive distribution utilities and village electrification programmes start yielding results. Ten states have joined the government's Ujwal Discom Assurance Yojna and the Centre plans to electrify 18,000 villages by May 2018. In Kerala most of the villages are going to be 100% electrified by 2020. Energy saving is the best method to produce energy. Most of the electronic appliances are now rated to meet the energy efficient standard. Most of the energy that produced is consumed by the domestic users and is consumed in terms of light. Energy efficient lighting like compact fluorescent lamps are hazardous to environment due to mercury contamination and are very difficult to recycle. LED lightings are highly efficient and make no hazard to environment and are the right choice for sustainable light energy source. Hence LED lightings are the need of the hour.

A LED lighting based skill development centre named as LeTin is proposed to give necessary training to students and electronics enthusiasts to make energy efficient lightings. The training for the aspirants will be given by the experts in the industry based on the MOU between LUTRON Technologies, Kochi (an Alumni venture) and the department of Electronics. The skill development centre will produce the LED lightings and solar based lighting solutions based on the training attained by the student is the final goal of this project

2. Needs/Problems

The basic need for the implementation of this project is to elevate this region into an energy efficient one. Even though there were initiatives by KSEB to distribute bulbs to its consumers, the quality and quantity were not met to address the necessary energy savings. Necessary awareness to the peoples around this area need to be given to use energy efficient lightings and appliances. In an around this college, three panchayats are targeted to give awareness in energy savings through LED lightings. Students are given training to present the factual data wise savings in the energy by using LEDs. Also they are equipped to present the impact of solar energy power which is going to revolutionize the energy sector.

3. Goals/Objectives

The desired goals and objectives are:

1. Memorandum of Understanding with industry for training students to assemble various LED modules
2. Training to around 20 selected students for setting LED and solar based lighting solutions.
3. Awareness to the public by students through house campaigning about the energy efficiency lightings.
4. Marketing of the finished LED light products through students with a reasonable profit to meet the financial constraints.
4. Development of standalone energy efficient solar light street lights and their marketing.

4. Procedures/Scope of Work

The first phase of this project envisaged to give in house training to students. The laboratories of the electronics department is equipped to meet the training by purchasing of soldering irons, multimeters, powermeters, luxmeters etc. About a week training is required to familiarize the various aspects in LED lightings. The training will be done by M/S LUTRON lightings, Muvattupuzha for about a week in designing and implementing the LED bulbs and tubes with standard power ratings. They will also give expertise training in solar based lightings to students.

5. Timetable

	Description of Work	Start and End Dates
Phase One	Training of selected students from UG programmes of B.Sc. Electronics by M/S Lutron Lightings on LED based lightings	11-2-2019 to 22-2-2019
Phase Two	Purchasing meters, soldering irons, LED bulb and Tube raw materials	Before 23-2-2019
Phase Three	Assembling the product, testing and marketing the product in the campus	1-3-2019 onwards

ACTIVITY	IMPLEMENTATION				RESPONSIBILITY
1. Design and implement training program for lighting solutions	Week 1	Week 2	Week 3	Week 4	CEO and staff, Lutron
2. Assembling the products					Staff, Lutron
3. Testing, Benchmarking the Product					Staff, Lutron, Faculty, Dept. of Electronics
3. Marketing the Product					Students
4. Improve procedure for project management					Head of the Department

6. Budget

	Description of Work	Start and End Dates
Phase One	Purchase of Soldering Iron, power meter and flux meter- ₹ 10,000	11-2-2019 to 22-2-2019
Phase Two	Purchase of LED bulb, Tube spare parts, Packing and labelling materials- ₹ 20,000	Before 23-2-2019
Phase Three	Marketing of the products- ₹ 1,000	1-3-2019
	Total expenses- ₹ 31,000	

The expenses may be met from the college resources with the prior sanction from the management as advance and may be paid back to the college funds after marketing the products to the college community.

7. Key Personnel

Persons who are involved in implementing the projects are:

Program coordinator -Dr. Praveen N, Associate Professor & Head, Department of Electronics

Sponsor - Mr. Shibu P L, CEO, Lutron Technologies, Kochi.

Team - Faculty:
Mrs. Baby Girija B, Assistant Professor of Electronics
Mr. Sunil Kumar K V, Associate Professor of Electronics
Dr. Saritha M, Associate Professor of Electronics
Dr. Reji A P, Assistant Professor of Electronics
Dr. Jinesh Mathew, Assistant Professor of Electronics
Dr. Jaison Peter, Assistant Professor of Electronics
Mr. Arun S Nair, Technical Assistant

Students:

1. Aravind M Ravi, S6 BSc Electronics
2. Akshay R Nair, S6 BSc Electronics
3. Aromal K S, S6 BSc Electronics
4. Ashilmon Shaji, S6 BSc Electronics
5. Anandhu Vijayakumar, S6 BSc Electronics
6. Bibin Alias, S6 BSc Electronics
7. Jaimon Ramanan, S6 BSc Electronics
8. Sreekutty Harikumar, S6 BSc Electronics

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